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Beach Profile Changes Due to Low Crested Breakwaters at Sigandu Beach, Central Java

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Abstract

Sigandu is a coastal tourism area, which is located in the coastal zone of Batang District. Central Java Province. This coastal area faces severe coastal erosion problem. It is indicated that erosion processes are occurred due to the jetty construction of Batang Commercial Port. This construction cause disturbance of longshore transport equilibrium. The objective of jetty construction is to trap the sediment, however, the classical problem of erosion and accration take place in a short time, soon after the jetty construction finished. The local government of Batang District becomes aware because of this erosion problem. Commenced on September 2014, the Marine and Fisheries Affair initiates the solution by applying low-crested breakwaters structures using geotextile tube. One month after the tube installation, rapid sedimentation occurred by the formation of salients and tombolos behind the structures. Beach profile calculation have been carried out through the series of monitoring and cross section measurement as well as visual observation. This paper presents results of monitoring beach activities. . The case study is expected to be an example and applied for other coastal areas having similar coastal problems.

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Keywords: low-crested breakwaters, geotextile tube, beach erosion, sedimentation rate, longshore drift.

1. Introduction

Sigandu beach is a sandy beach which is located in Batang District, Central Java Province. This coastal area is flanked by two perpendicular structures stretched seaward. In the west side the jetty of Batang Commercial Port

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which has a length of 250 m. In the eastern part, the reclamation is already conducted with approximately 150 m seaward. The destruction of Sigandu beach is indicated due to the jetty construction. The effect of jetty construction leads to the accretion on the updrift area and erosion on the down drift area since this jetty block the longshore transport process (see Fig. 1). In addition, it is indicated that reclamation area in the eastern part is also contributed in the erosion process.

The beach experiences the tide which is semi-diurnal in nature with a tidal range of 0.90 m. Flood tide moves from north to south whereas ebb tide moves in opposite direction and tidal movement is parallel to the coast. The project site is located on a gentle topography which beach slope varying between 0-2%. Meanwhile the bathymetry of the coast varying between 0-3 m on the beach and the deepest of 10 m (Fig.2. Normal wind speed is between 1 and 18.5 m/s with a predominant wind occurs during west monsoon. The project site is affected mainly by monsoon generated waves. Average wave height is varying from 0.15 m to 1.8 m with dominant wave directions of 280ºN to 320ºN during west monsoon and 45^0N to 90^0 N during east monsoon (Figure 3). The objective of this paper is to evaluate the effectiveness of Geotube LCB performance to overcome the erosion problem in the Sigandu Beach. In addition, the technical advice of rehabilitation and avoid the continuity of erosion rate. The activity scope of this study is collecting data and information, visual analysis, measuring the beach profile and current, data analysis, and technical advice.

Nomenclature

LCB	Low Crested Breakwater
h	height of the structure
d	water depth

2. Theoretical Framework

2.1. LCB Concept

LCB is to be developed by cutting the structure height of conventional non-overtopping breakwaters, which is directly proportional to the volume and cost reduction. Its protection capacity is as effective as conventional emerging breakwaters. The LCB crest close to the still water level or slightly higher, and the main objective of reducing their energy in the coastal area behind it (Buccino and Calabrese, 2007). Therefore, as the name implies, LCB is a coastal protection structures that are always overtopped by the wave. Moreover, the structure may have positive or negative freeboard and the armor layer is attacked by the breaking waves (Pascual et al., 2007). The LCB structures are constructed in such a way that the waves breaking and overtopping of the structure and causing wave energy dissipated. When the wave breaks over the structure, a certain amount of water is likely to be pumped over it. The net discharge across the structure creates an accumulates mass in the lee side, behind the structure which establishes a higher mean water level, referred to as piling-up. In a serial or discontinuous low-crested breakwater, i.e. serial breakwaters that are interrupted by gaps, piling-up may generate strong currents through the gaps. These currents are called as rip currents, they are responsible of scouring at the toe of the structures, resulting in severe erosion along the gap, and may be dangerous to swimmers (Bellotti, 2004; Lamberti et.al, 2007).



Figure 1. Location of Sigandu Beach, District of Batang, Central Java (Google Earth, 2015)

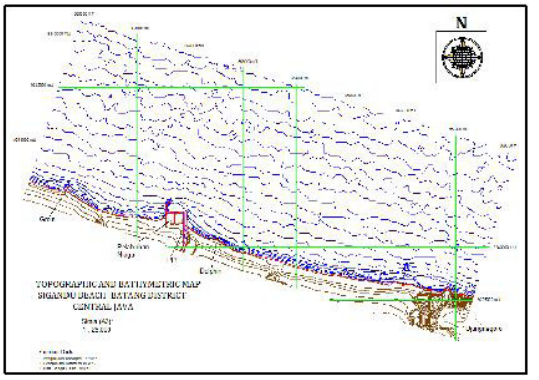


Figure 2. Topographic and bathymetric map of Sigandu beach (DKP, 2014)

The LCB structure will be functioning properly if its existence is not only able to restore eroded beaches behind the structure, but also able to widen the shore seaward with the formation of salient and tombolos. Similarly, the waves energy that pass through over the structure will encourage sediment to accumulated on the beach. LCB constructed using a gap in between, in terms of the pattern of the dynamics of the mass of water to give a physical impact as follows: 1) reducing the piling-up behind the structure. Water mass accumulated behind the LCB, will flow back towards offshore through the gap, therefore reducing the piling-up; 2) decreasing wave reflection effects. Wave reflection due to the existance of LCB causes instability of the structure by scouring at the toe of the structure seaward. However, the LCB structure installed in between, at the same time, generate strong currents in the gap which called as rip currents. This strong water mass flow serves to reduce the piling-up behind the structures. The flow velocity of the strong rip currents, potentially eroding and transporting coastal sediments behind the LCB's. In addition to causing coastal erosion behind the structure, strong currents can also cause scour at the end of the LCBs in the gap, which can threaten the stability of the structure.

The flow velocity in the gap decreases exponentially with increasing gap width (Sulaiman et. al., 2014). Therefore, for practical planning purposes and to get more effective low-crested breakwaters, it requires an optimal gap width, which is able to minimize the piling-up, but also do not cause strong currents which can cause unwanted effects.

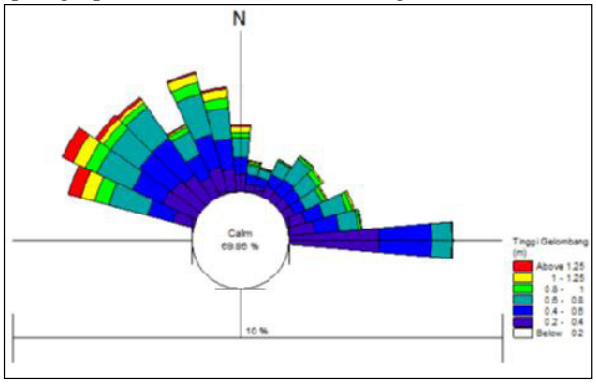


Figure 3. Waverose of Batang 2006-2013 (DKP, 2014)

2.2. Determination of the LCB Freeboard

The main parameters used in describing the LCB geometry shown in Figure 4. In this case h = height of the structure, d = depth of water, and Rc = h-d is freeboard, which is the difference between the crown level of structures and still water level. One of the important parameters in determining the effectiveness of the breakwater is submergence degree of the LCB, which can be explained by three parameters, namely: (1) the degree of submergence or freeboard, Rc = h - d; (2) relative height of the structure = h/d; and (3) the ratio between freeboard and water depth= Rc/d. Submergence degree is the ratio between the water depth of the structure height of the LCB. For emerging breakwater, with crown level beyond the depth of water, the ratio is less than one. Relatively high structure, which is the ratio between the heights of the structure to a depth of water (h/d) can also be used as a non-dimensional parameter to describe the degree of submergence and the degree of exposure.

The freeboard, defined as the difference in height between elevation peak water level structures with an average:

$$R_c = h - d \quad (1)$$

Where R_c is freeboard, h is the structure height, and d is water depth in front of the structure. By Equation (1) the positive freeboard obtained for the emerging breakwaters and negative freeboard for the LCB's. Therefore, freeboard for LCB's can be referred to as the degree of submerged. Non-dimensional parameters for relative freeboard that is a freeboard ratio which means the ratio between the degree of submergence and water depth:

$$\frac{Rc}{d} = \frac{(h-d)}{d} = \frac{h}{d} - 1 \quad (2)$$

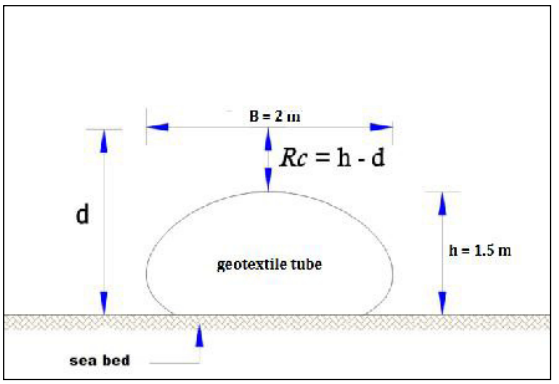


Figure 4. Geometric parameter of LCB structure (Sulaiman et.al., 2013)

2.3 Use of Geotextile

Various construction materials may be used for coastal protection structures ranging from the conventional (natural rocks and concrete blocks) until the leading edge of innovation outcomes such as geotextiles and gabion. Among the materials the results of these innovations, geotextile is the most popular material as material protection structure. An alternative material for coastal protection which have softer and cheaper such as sandbags geotextile (geotextile sand containers, GSC) or in the market known as Geotube, widely used as a substitute for conventional materials are expensive and rigid (Oumeraci and RECIO, 2010).

Geotextile tube is a big sack (width and height of approximately 2 m and 1.5 m) geotextiles made from woven or non-woven, shaped like a bolster pillow filled with sand with a length of 20 m, and is used for coastal protection and river (Fig. 5). Geotextile is now a substitute material for natural rockws has increasingly known in protecting coastal erosion. This method of Coastal protection use geotextile tube material for offshore breakwaters, as substitute material of natural rocks. The use of geotextile provides several advantages, namely: 1) The geotextile material such as Geotube or Geotextile Sand Containers are available in the market with various sizes; 2) relatively lightweight geotextile material which allows freight to a location relatively inexpensive; 3) a filler material such as sand obtained at the project site; 4) a flexible response to the wave; and 5) can be installed by using a local power without the need for special expertise.

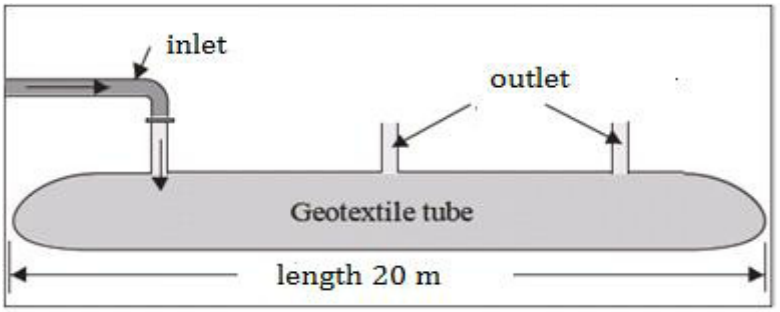


Figure 5. Typical geotube LCB (Sulaiman et.al., 2012)

Table 1. Specfication for Geotextile Tube Application (DKP, 2014)

Properties	Unit
Tube Circumference	Antara: 2.3 m to 18 m
Kuat Tarik (Tensile)	Min : 175 kN X 175 kN / m
Kemuluran (Elongation)	Max : 15% X 15%
Kuat Sobek Trapezoidal (Tear)	Min : 2.7 kN X 2.7 kN
Kuat Tusuk (Puncture)	Min : 1.8 kN
Kuat Jahitan (Seam)	Min : 60% of Tensile
Ukuran Bukaan yg Tampak (AOS)	Max : 0.425 mm
Volume Aliran Air (Flow rate)	Min : 240 lt/minute/m2

3. The Instalation of LCB with Geotube

3.1. Spesification and Properties of Geotextile

Geotextile material used as an ingredient in Geotube have high tensile strength so as to withstand the force and pressure during the filling process and after Geotube fully charged. In addition, this material also should have the size of the opening and optimum permeability so that water can quickly get out but able to withstand stuffing sand does not come out. Geosynthetic ultimate tensile strength required for 200kn/m. Geotextile tube filler materials such as sand and are assumed to have weight (γ) of 18kN/m3. Geotextile material is geosynthetic fibers produced by needle-punch. The raw material is made of polyester material (PES)/Polypropylene colored fibers. Density of the material as a whole must be greater than 1 (one). The type of geotextile material for the fabric tube is a kind of non-woven high strength, whether it is monofilament, multifilament or fibrillation. The type of polymer used to make the yarn is not specifically defined, but based on the existing production process today usually use polyester yarn and polypropylene yarn. The end result of non-woven material must have the physical characteristics / data as shown below.

3.2. Layout and Gap Width Optimalization of Geotube LCB

A total of 14 units of Geotube LCBs has been installed in counter-measuring effort of beach erosion problems along Sigandu beach. Formation of initial installation consists of 3 gaps with a gap width of 20 m as shown in Figure 6. Geotube LCBs of Sigandu beach were arranged serially with the composition of: 5 Geotubes on the left side (in the West side), 5 Geotubes in the middle, and 5 Geotubes on the right side (East Side). The distance from the coast (L_x) to the position of LCBs is approximately 75 m. The position and the distance of Geotube LCBs to the coastline are able to allow the formation of tombolos. While the LCB's position to tidal fluctuations are installed at a mean sea level on average (MSL) as shown in Figure 7.

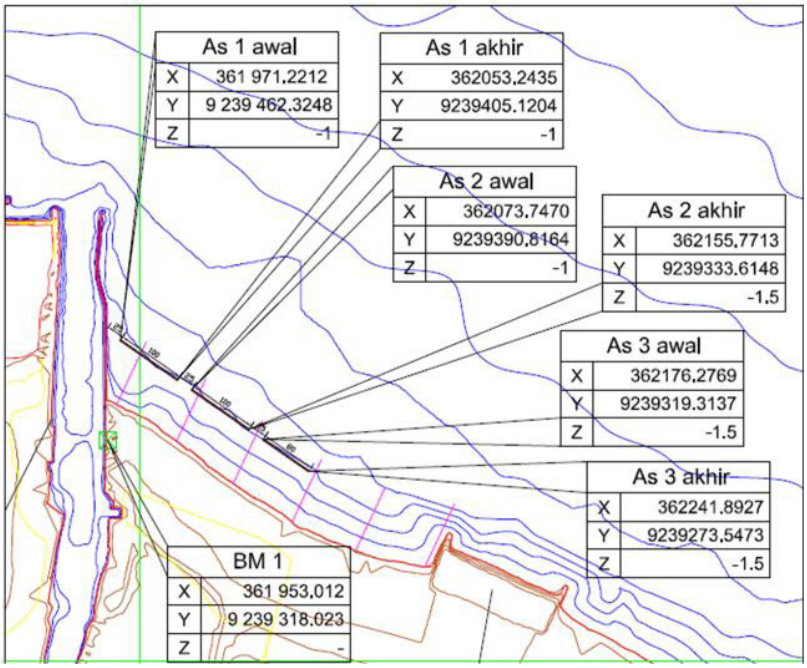


Figure 6. Layout of geotube LCB installation on Sigandu Beach (DKP, 2014)

The objective of geotube LCBs installation is planned to reduce severe coastal erosion and encourage the accumulation of sediments, thus forming a new beach. Practical approaches to generate an effective LCB as erosion control measure is to place the geotube LCBs at an elevation above MSL or below the highest water level (Fig. 7). Geotube LCBs constructed along Sigandu beach provide a positive response to the formation of salient and tombolos. An accreted sand form a new beach behind the LCBs.

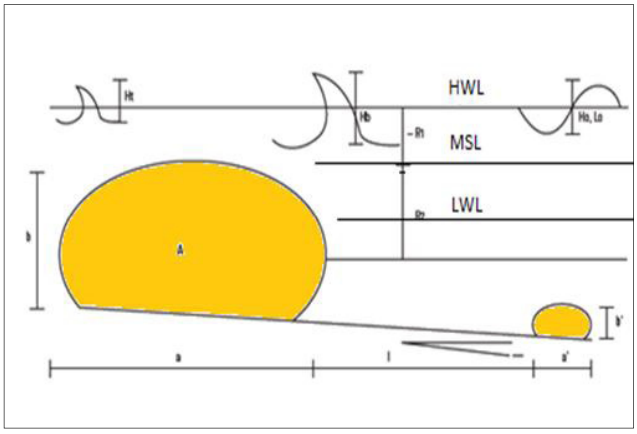


Figure 7. Placement of Geotube LCB above or at MSL (Sulaiman et.al., 2012)

4. Results and Discussions

4.1. Beach Profile Changing

Changes of Sigandu coastline can be seen visually with increasing accreted sediment behind Geotube LCBs. The advance of the coastline and the widening of the coast seaward approach the LCB structure, occurs along the beach installed by the LCBs. The accumulation of sediment that make up a new coastline in the form of salient, visible along the beach behind the LCB. In some spots and points of measurement, has formed some tombolos, a new coastline due to sediment accretion that almost approach the LCB structures. Photographs of shoreline change before and after the LCB installation are presented in Figure 8. To get more information regarding how much changes after the installation of the tube, beach profile measurement has been conducted along the coast from the jetty eastward. Beach profile measurement performed every 25 m.

4.2. Estimation of Accreted Sediment on the Beach

From the analysis of beach profile measurement results on 7th November, 2014, it is identified that approximately 1 month after the LCBs installed in Sigandu Beach, the amount of sediment accreted behind the LCBs is in average of 17,627 m^3. Which the beach length of 280 m and its width of 75 m, so that the thickness of accreted sediment is around of 84 cm.

The source of the sediment that accumulated on the beach behind geotube LCB comes from offshore-onshore sediment transport and longshore sediment transport. In addition, the advantage of the LCB and serial LCB's compared to conventional emerging breakwater is that sediment transport towards the coast takes place continuously all the time, either during high tides and low tide. At the time of a lower sea level, sediment may able to enter the beach through a gap in between the LCB's.

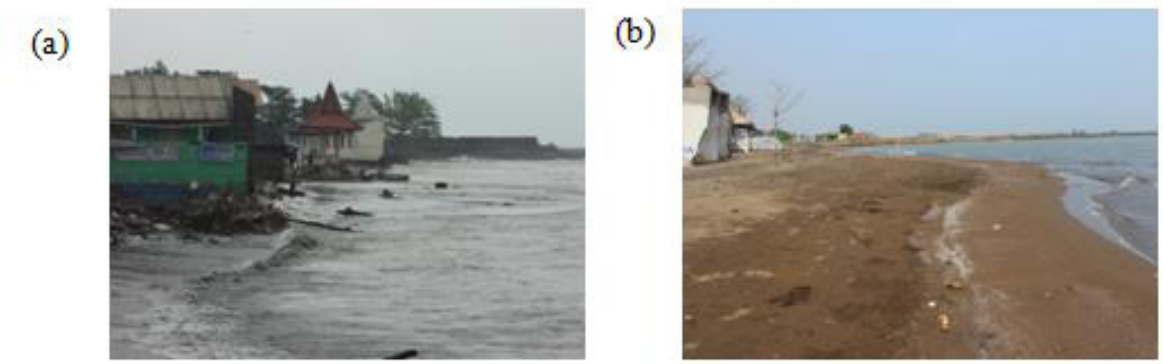


Figure 8. Sigandu Beach condition (a) before and (b) after LCB installation.

The presence of Sigandu beach situated on the upstream of Batang Commercial Port, during the east monsoon the direction of the dominant sediment transport is coming from East to West. The jetty of Batang Niaga Port at the East monsoon plays an important role in blocking of longshore sediment transport that coming from the east side and the entire sediment transport westward accrated and halted by the jetty. Therefore, the existence of the Niaga Batang Port is the most probably expected to become a major cause rapid accumulation of sediment after the LCB Geotube installation.

Another source of sediments that accelerate the development of the coastline Sigandu is an abundance of sediment in front of geotube LCBs, most likely derived from the results of dredging the harbor to the west coast of Sigandu beach. Sediment that is transported by waves and currents over the LCB structure. The estimate is quite

reasonable considering the results of the observation showed that dredging of the port rod, partially discharged into the beach area in front of the LCB Geotube.

4.3. Performance and condition of LCB with Geotube

Geotube LCBs installed since October 2014 as a whole showed a good condition. Geotube position at some point experienced settlement, it appears at low tide with the surface of the tubes identify partially emerging and submerging (See Fig. 9). Oblique position of some tubes are identified at the gap between the LCBs, both edges of the left and right ends. This slanting is predicted due to the occurrence of scour at the toe end of tubes as a result of strong currents in the gap that referred to rip currents. The strong currents arising in the gap stems from unproporsional gap width compared to the length of the tubes. Reverse flow toward the sea comes from the accumulation of water mass behind submerged structure, known as the piling-up. Heaps of water masses are then turned toward the sea through the gap and generate a strong return flow. The flow velocity depends on the gap width. A narrower gap generate stronger currents. Therefore optimizing the width of the gap needs to be determined at the planning stage. Other toe scouring occurs in the toes of the tubes shoreward, this condition is indicated by deepening of water depth behind the structures.

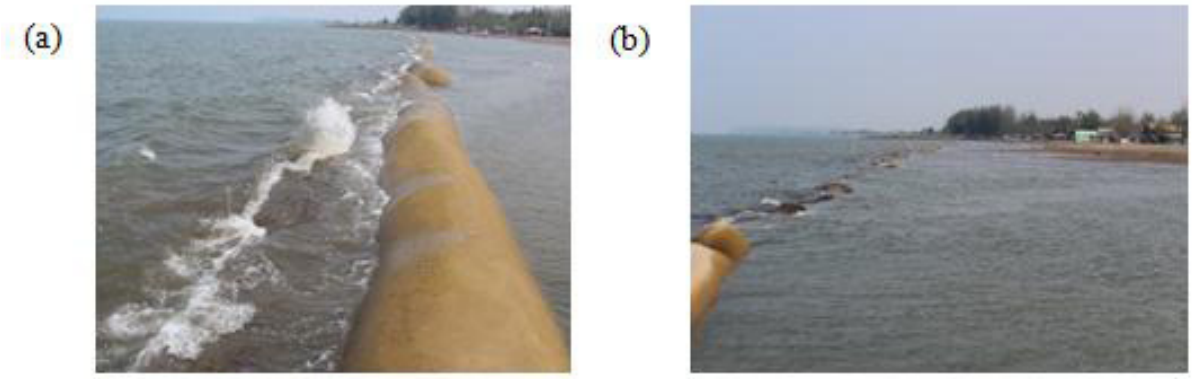


Figure 9. Condition of geotube LCB of Sigandu Beach

5. Conclusions and Recommendation

5.1. Conclusions

- The pressence of the Batang Commercial Port plays an important role in seasonal changes of Sigandu beach. The port act as a dam that traps longshore sediment transport.
- Counter-measuring coastal erosion along Sigandu beach by installing geotube LCBs has shown remarkable results. Previously eroded shoreline, in a relatively short time has established a new coastline by formation salient and tombolos.
- Placement of the LCB's position, the length of structure, and the gap width between the LCBs determine the successful of the LCB's application.
- Serial geotube LCBs play a major role in the process of sedimentation, transport of sediment into the beach will continue through the gap even during low tide.
- Geotube LCBs are designed as coastal protection efforts that can generate beach profile balance. Using alternative materials such as geosynthetic has contributed to the coastal areas which lack of the resources such as an adequate natural rocks.
- The growing demand for tourist beaches and beaches aesthetic situation contributes greatly to the application of LCBs as environmentally friendly, due to reduce visual effects.

5.2 Recommendations

- Some improvements need to be made to the structure geotube LCBs include widening of the gap, in order that the resulting flow is not too large impact on the local scour at the toe.
- Due to material geotextile tube is fairly susceptible and attract the attention of tourists, special care must be made such as warning signs and security assigned to protect it.
- Rehabilitation of Sigandu beach through the installation of geotube LCBs, will be more effective when combined with the planting of coastal vegetation, such as mangroves that will act as a binder of sediments and natural protector against extreme conditions.
- Protection of Sigandu beach on the east side should be carried out sooner in order to establish a single system of Sigandu beach which will tend to be more stable and avoid the influence of reflection waves and currents from adjacent reclamation land.
- The successful effort in countermeasuring erosion problems of Sigandu beach, needs to be disseminated for public references.

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